

Saurav Raj Kunwar

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Professional Profile

Applied microeconomist specializing in productivity, firm performance, causal inference, structural modeling, and decision-making under uncertainty. Develops empirical frameworks to measure performance, quantify economic tradeoffs, and evaluate how incentives and external conditions shape outcomes. Research spans firm productivity, resource allocation, valuation, innovation, and decision analysis using complex observational, panel, spatial, and administrative data.

Selected Economic Research Projects

Did Merger Remedies Preserve Competition in U.S. Cottonseed Markets? A Structural Retrospective of Bayer-Monsanto

Research Project

- Develop a structural differentiated-products framework using PyBLP to model cottonseed demand, firm ownership, pricing, and market power under alternative merger and divestiture scenarios.
- Design a retrospective merger analysis comparing pre-merger competition, the observed BASF divestiture, and a no-remedy counterfactual, with intended outcomes including prices, market shares, markups, grower surplus, and firm profits.

Index Investing and Productivity of U.S. Firms

Working paper in progress

- Apply difference-in-differences and triple-difference designs to evaluate how index investing relates to productivity outcomes among U.S. firms.
- Examine how capital-market incentives and firm behavior relate to changes in productivity and performance over time.

Agricultural TFP, Public R&D, and Patents

- Analyze long-run productivity convergence across U.S. states and the contribution of public R&D and innovation to productivity growth.
- Link research investment, patent activity, and productivity performance to study how innovation contributes to differences in long-run economic outcomes.

Firm Efficiency, Productivity, and Economic Tradeoffs

Dissertation and published research

- Develop structural and econometric models to measure firm and producer performance under resource, environmental, and uncertainty constraints.
- Estimate efficiency measures and shadow prices to quantify economic tradeoffs and inform resource-allocation and policy decisions.

Weather Risk, Resource Misallocation, and Smallholder Productivity

Solo-authored research in progress

- Integrate household panel data with satellite-based climate measures to study how major weather shocks affect productivity and resource allocation.
- Apply production-function estimation, productivity decomposition, and event-study methods to measure changes in producer performance following external shocks.

Economic Value of Irrigation Information for Texas High Plains Cotton

USDA-NRCS Conservation Innovation Grant project

- Develop an expected-value-of-information framework to quantify the economic value of improved information for management decisions.
- Contribute to the design of a randomized on-farm experiment evaluating how information affects resource allocation and economic outcomes under uncertainty.

Professional Experience

Postdoctoral Researcher

Texas Tech University, Department of Agricultural and Applied Economics

2026–Present

Lubbock, TX

- Conduct applied economic research on productivity, performance measurement, producer behavior, innovation, and decision-making under uncertainty.

- Lead and support grant development by framing research questions, designing empirical strategies, coordinating collaborators, and translating technical concepts into fundable proposals.
- Build empirical workflows that connect economic theory, quantitative methods, and complex data to decision-relevant economic interpretation.
- Collaborate across projects involving productivity, innovation, resource allocation, and the economic value of information.

Research Scholar

2023

University of Maryland, Department of Agricultural and Resource Economics

College Park, MD

- Collaborated with Robert G. Chambers on production economics, firm efficiency, environmental performance, and structural economic modeling.
- Developed research linking econometric estimation, productivity measurement, and policy-relevant performance metrics.

Graduate Research Assistant

2020–2026

University of Illinois Urbana-Champaign

Urbana, IL

- Conducted empirical research on productivity, environmental valuation, conservation, and producer behavior using field-level and panel data.
- Built reproducible empirical workflows supporting peer-reviewed research, working papers, and applied decision-support analysis.

Education

University of Illinois Urbana-Champaign

2026

PhD, Agricultural and Applied Economics

Dissertation: Three Essays on Environmental Services Valuation and Agricultural Productivity

University of Georgia

2020

MS, Agricultural and Applied Economics

Tribhuvan University

2017

BS, Agriculture

Selected Publications

- Kunwar, S. R., Chambers, R. G., Gentry, L., & Serra, T. (2026). “What Are the Carbon Services from Cover-Crop Adoption Worth from Farmers’ Perspective?” *American Journal of Agricultural Economics*, 108(3), 901–925.
- Chambers, R. G., Kunwar, S. R., Quiggin, J., Serra, T., & Wang, S. (2025). “Estimating Production and Environmental Efficiency under Uncertainty.” *Annual Review of Resource Economics*, 17, 85–104.
- Kunwar, S. R., Fonsah, E. G., & Escalante, C. L. (2025). “An Assessment of Profitability Using Monte Carlo Simulation Approach: A Case of Georgia Blueberry Industry.” *Journal of Food Distribution Research*.

Selected Recognition

- Anthony Grano Fellowship, Agricultural and Applied Economics Association, 2025.
- Selected Representative, Catalyzing Advocacy in Science and Engineering (CASE), American Association for the Advancement of Science, Washington, D.C., 2024.
- Selected Participant, Global Young Scientists Summit, 2023.

Technical Skills

Econometrics: Difference-in-differences, triple-difference designs, event studies, instrumental variables, panel methods, spatial econometrics, control-function estimators, system GMM.

Economic Modeling: Structural modeling, production-function estimation, stochastic frontier analysis, productivity and efficiency measurement, shadow pricing, simulation, mathematical programming, and PyBLP.

Applications: Firm productivity, performance measurement, resource allocation, valuation, incentives, innovation, policy and program evaluation, decision-making under uncertainty.

Programming and Data: R, Python, Stata, GAMS, SAS, SPSS; firm-level, panel, household, administrative, patent, geospatial, and remote-sensing data.

Spatial and Research Tools: Google Earth Engine, QGIS, Git/GitHub, LaTeX; reproducible and version-controlled analytical workflows.

Languages: English, Nepali, Hindi.